Computer Networks
Assignment - 6
problems

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Date:

P1 two dim parity scheme 1110 0110 1001 0101

arrange data into 4x4 grid

calculate (*) even parity (row) bits

r_1	1110	1	1+1+1+0 = 3	odd ->	1
r_2	0110	0	0+1+1+0 = 2	even ->	0
r_3	1001	0	1+0+0+1 = 2		0
r_4	0101	0	0+1+0+1 = 2		0
	0100	1			

even parity column bits

1+0+1+0 = 2 0

1+1+0+1 = 3 1

1+1+0+0 = 2 0

0011 ~~[illegible]~~ = 2 = 0

1+0+1+0 = 1 1

find matrix

1110 1
0110 0
1001 0
0101 0
0100 1

PS $G = 10011$ Sb.Ls Polynomial $x^4 + 1 + x^3 + 0 + x^2 + 0 + x^1 + 1 + x^0 + 1$

$D = 1010101010$ $\rightarrow x^4 + x^1 + 1$

Deg $\rightarrow 4$

$r = 4$

$D + r$ $1010101010\boxed{00\ 00}$

$R =$

```
          10110110
10011) 1010 1010 1000 00
       10011
        1100
        0000
        11001
        10011
         10400
         10011
          1111
          0000
           11110
           10011
            11010
            10011
             10010
             10011
               010
               000
               0100
               0000
               0100
```

therefore, the value of $R = 0100$.

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P6 a) D = 1000 1001 0100 00

```
          100100 1000
10011 ) 1000 1001 0100 00
        10011
         10011
          10011
           10010
           10011
             1000
```

R = 100

b) D = 01011 01010 0000

```
          0101010101
10011 ) 0101101010 0000
        10011
         10110
         10011
          10100
          10011
           11100
           10011
            1111
```

R = 1111

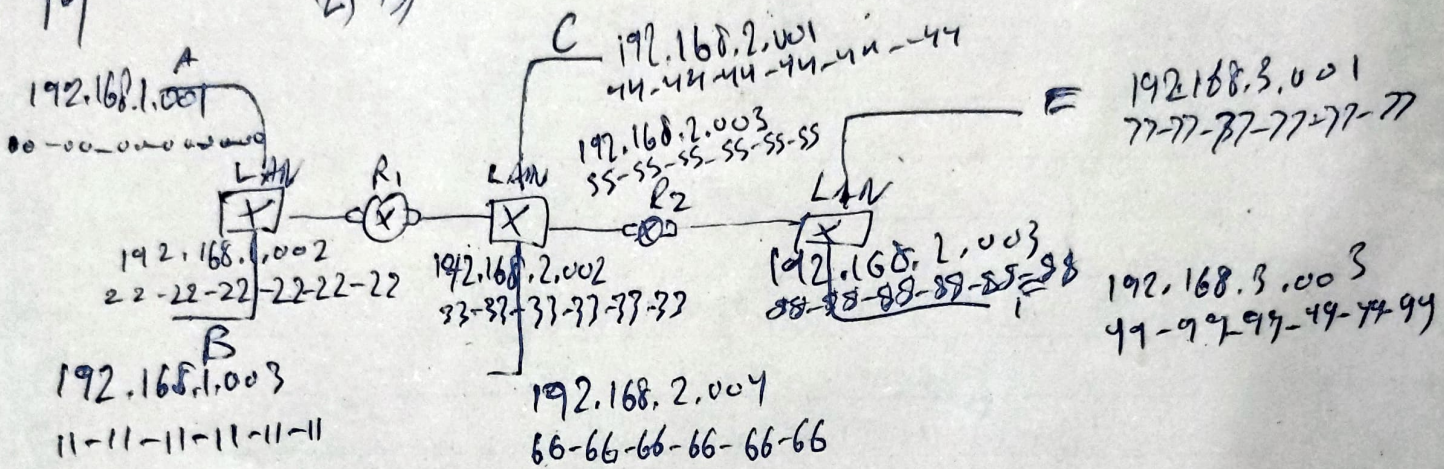
c) D = 0110100011 0000

```
                 011000 1010
10011 ) 0110100011 0000
        10011
         10010
         10011
          10110
          10011
           101 00
               0011
               1110
```

R = 1110

P14

2) b)



c) E → B

1 forwarding table in E determines datagram routed to net 192.168.2.003

2 The adapter in E creates ethernet packet with dest address 88-88-88-88-88-88

3 R2 receives packet and extracts datagram forwarding table indicates ~~router~~ datagram to 192.168.2.002

4 R2 sends ethernet packet with dest address of 33-33-33-33-33-33 and source address 55-55-55-55-55-55 via interface 192.168.2.003

5 Process continues till packet reaches dest B

d) ARP in E queries MAC address of 192.168.2.002. E sends an ARP query packet within broadcast ethernet frame. R2 receives query packet and sends back an ARP response packet. ARP response packet is carried by ethernet frame with dest 77-77-77-77-77-77

Date:

	Src MAC add	Dest MAC add	Src IP	Dest IP
P12) i) from A to SW :	00-00-00-00-00-00	55-55-55-55-55-55	111.111.111.001	133.333.333.003
ii) from Sw to right router:	00-00-00-00-00-00	55-55-55-55-55-55	111.111.111.001	120.222.222.002
iii) from right router to F :	11-00-00-00-00-00	99-99-99-99-99-99	111.111.111.001	132.332.332.003

P16)

i)

B sends frame to E

Switch table state = Switch learns int corresponding to MAC add of B

Links packet is forwarded to A, B, C, D, E, F

Exp switch table is empty so switch learns only just the int corresponding to MAC add of E

ii)

E replies with frame to B

- Switch learns int corresponding to MAC add of E
- B usage switch already knows int corr to B

iii) A sends a frame to B

- Sw learns the interface corr to A ; B ; since origin

iv) B replies with frame to A

- Sw table stays same as before ; A ; already learnt int corresponding to MAC add of A.